

REFERENCES

- [Abo03] H. Abou-Kandil, G. Freiling, V. Ionescu, and G. Jank, *Matrix Riccati Equations in Control and Systems Theory*, Birkhauser, Boston, Massachusetts, 2003.
- [Aid83] V. Aidala and S. Hammel, "Utilization of modified polar coordinates for bearings-only tracking," *IEEE Transactions on Automatic Control*, **AC-28**(3), pp. 283-294 (March 1983).
- [Aki03] B. Akin, U. Orguner, and A. Ersak, "State estimation of induction motor using unscented Kalman filter," *Conference on Control Applications*, pp. 915-919 (June 2003).
- [Ale91] H. Alexander, "State estimation for distributed systems with sensing delay," *Proceedings of the SPIE, Data Structures and Target Classification*, **1470**, pp. 103-111 (1991).
- [Als72] D. Alspach and H. Sorenson, "Nonlinear Bayesian estimation using Gaussian sum approximations," *IEEE Transactions on Automatic Control*, **AC-17**(4), pp. 439-448 (August 1972).
- [Als74] D. Alspach, "Gaussian sum approximations in nonlinear filtering and control," *Information Sciences*, **7**, pp. 271-290 (1974).
- [Als74b] D. Alspach, "A parallel filtering algorithm for linear systems with unknown time varying noise statistics," *IEEE Transactions on Automatic Control*, **AC-19**(5), pp. 552-556 (October 1974).
- [And79] B. Anderson and J. Moore, *Optimal Filtering*, Prentice-Hall, Englewood Cliffs, New Jersey, 1979.
- [And68] A. Andrews, "A square root formulation of the Kalman covariance equations," *AIAA Journal*, **6**(6), pp. 1165-1166 (June 1968).

- [And04] C. Andrieu, A. Doucet, S. Singh, and V. Tadic, "Particle methods for change detection, system identification, and control," *Proceedings of the IEEE*, **92**(3), pp. 42-438 (March 2004).
- [Ant91] A. Antoulas, *Mathematical System Theory*, Springer-Verlag, New York, 1991.
- [Aok65] M. Aoki, "Optimal Bayesian and min-max control of a class of stochastic and adaptive dynamic systems," *Symposium on Systems Engineering for Control System Design*, pp. 77-84 (1965).
- [Aok67] M. Aoki, *Optimization of Stochastic Systems*, Academic Press, New York, 1967.
- [Ara94] J. Aranda, J. De La Cruz, S. Dormido, P. Ruiperez, and R. Hernandez, "Reduced-order Kalman filter for alignment," *Cybernetics and Systems*, **25**, pp. 1-16 (1994).
- [Aru02] M. Arulampalam, S. Maskell, N. Gordon, and T. Clapp, "A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking," *IEEE Transactions on Signal Processing*, **50**(2), pp. 174-188 (February 2002).
- [Ath68] M. Athans, R. Wishner, and A. Bertolini, "Suboptimal state estimation for continuous-time nonlinear systems from discrete measurements," *IEEE Transactions on Automatic Control*, **AC-13**(5), pp. 504-514 (October 1968).
- [Ath77] M. Athans, R. Whiting, and M. Gruber, "A suboptimal estimation algorithm with probabilistic editing for false measurements with applications to target tracking with wake phenomena," *IEEE Transactions on Automatic Control*, **AC-22**(3), pp. 372-384 (June 1977).
- [Atk89] K. Atkinson, *An Introduction to Numerical Analysis*, John Wiley & Sons, New York, 1989.
- [Bai95] E. Bai, K. Nagpal, and R. Tempo, "Bounded error parameter estimation: Noise models, recursive algorithms and H_∞ optimality," *American Control Conference*, pp. 3065-3069, June 1995.
- [Bal87] A. Balakrishnan, *Kalman Filtering Theory*, Optimization Software, New York, 1987.
- [Bal01] J. Ballabrera-Poy, A. Busalacchi, and R. Murtugudde, "Application of a reduced-order Kalman filter to initialize a coupled atmosphere-ocean model: Impact on the prediction of El Nino," *Journal of Climate*, **14**, pp. 1720-1737 (April 2001).
- [Ban91] R. Banavar and J. Speyer, "A linear-quadratic game approach to estimation and smoothing," *American Control Conference*, pp. 2818-2822 (1991).
- [Ban92] R. Banavar, "A game theoretic approach to linear dynamic estimation," Doctoral Dissertation, University of Texas at Austin, May 1992.
- [Bar83] I. Bar-Itzhack and Y. Medan, "Efficient square root algorithm for measurement update in Kalman filtering," *Journal of Guidance, Control, and Dynamics*, **6**(3), pp. 129-134 (May 1983).
- [Bar78] Y. Bar-Shalom, "Tracking methods in a multitarget environment," *IEEE Transactions on Automatic Control*, **AC-23**(4), pp. 618-626 (August 1978).
- [Bar95] Y. Bar-Shalom and X. Li, *Multitarget-Multisensor Tracking: Principles and Techniques*, YBS Publishing, Storrs, Connecticut, 1995.
- [Bar98] Y. Bar-Shalom and X. Li, *Estimation and Tracking: Principles, Techniques, and Software*, YBS Publishing, Storrs, Connecticut, 1998.
- [Bar01] Y. Bar-Shalom, X. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*, John Wiley & Sons, New York, 2001.

- [Bar02] Y. Bar-Shalom, "Update with out-of-sequence measurements in tracking: Exact solution," *IEEE Transactions on Aerospace and Electronic Systems*, **38**(3), pp. 769-778 (July 2002).
- [Bar05] K. Barbosa, C. de Souza, and A. Trofino, "Robust H_2 filtering for uncertain linear systems: LMI based methods with parametric Lyapunov functions," *Systems and Control Letters*, **54**(3), pp. 251-262 (March 2005).
- [Bas99] O. Bas, B. Shafai, and S. Linder, "Design of optimal gains for the proportional integral Kalman filter with application to single particle tracking," *IEEE Conference on Decision and Control*, pp. 4567-4571 (December 1999).
- [Bas66] R. Bass, V. Norum, and L. Swartz, "Optimal multichannel nonlinear filtering," *Journal of Mathematical Analysis and Applications*, **16**(1), pp. 152-164 (1966).
- [Bat62] R. Battin, "A statistical optimizing navigation procedure for space flight," *American Rocket Society Journal*, **32**, pp. 1681-1696 (November 1962).
- [Bat64] R. Battin, *Astronautical Guidance*, McGraw-Hill, New York, 1964.
- [Bat82] R. Battin, "Space guidance evolution - A personal narrative," *Journal of Guidance*, **5**(2), pp. 97-110 (March-April 1982).
- [Bay99] J. Bay, *Fundamentals of Linear State Space Systems*, McGraw-Hill, New York, 1999.
- [Bel67] J. Bellantoni and K. Dodge, "A square root formulation of the Kalman-Schmidt filter," *AIAA Journal*, **5**, pp. 1309-1314, 1967.
- [Bel60] R. Bellman, *Introduction to Matrix Analysis*, McGraw-Hill, New York, 1960.
- [Bel61] R. Bellman, *Adaptive Control Processes: A Guided Tour*, Princeton University Press, Princeton, New Jersey, 1961.
- [Bel80] R. Bellman, "Some consequences of the nonnegativity of the elements of the matrix exponential," *Nonlinear Analysis, Theory, Methods and Applications*, **4**(4), pp. 735-736 (1980).
- [Ben62] T. Benedict and G. Bordner, "Synthesis of an optimal set of radar track-while-scan smoothing equations," *IRE Transactions on Automatic Control*, **AC-7**(4), pp. 27-32 (July 1962).
- [Ber85] D. Bernstein and D. Hyland, "The optimal projection equations for reduced-order state estimation," *IEEE Transactions on Automatic Control*, **AC-30**(6), pp. 583-585 (June 1985).
- [Ber89] D. Bernstein and W. Haddad, "Steady-state Kalman filtering with an H_∞ error bound," *Systems and Control Letters*, **12**, pp. 9-16 (1989).
- [Ber05] D. Bernstein, *Matrix Mathematics*, Princeton University Press, Princeton, New Jersey, 2005.
- [Bet94] M. Bettayeb and D. Kavranoglu, "Reduced order H_∞ filtering," *American Control Conference*, pp. 1884-1888 (June 1994).
- [Bie73] G. Bierman, "A comparison of discrete linear filtering algorithms," *IEEE Transactions on Aerospace and Electronic Systems*, **AES-9**(1), pp. 28-37 (January 1973).
- [Bie76] G. Bierman, "Measurement updating using the U - D factorization," *Automatica*, **12**(4), pp. 375-382 (July 1976).
- [Bie77a] G. Bierman and C. Thornton, "Numerical comparison of Kalman filter algorithms: Orbit determination case study," *Automatica*, **13**(1), pp. 23-35 (January 1977).

- [Bie77b] G. Bierman, *Factorization Methods for Discrete Sequential Estimation*, Academic Press, San Diego, California, 1977.
- [Bil95] P. Billingsley, *Probability and Measure*, Wiley-Interscience, New York, 1995.
- [Bit85] R. Bitmead, M. Gevers, I. Petersen, and R. Kaye, "Monotonicity and stabilizability properties of solutions of the Riccati difference equation: Propositions, lemmas, theorems, fallacious conjectures and counterexamples," *Systems and Control Letters*, **5**(5), pp. 309-315 (April 1985).
- [Bit91] S. Bittanti, A. Laub, and J. Willems (eds.), *The Riccati Equation*, Springer-Verlag, New York, 1991.
- [Bjo67] A. Bjorck, "Solving linear least squares problems by Gram-Schmidt orthogonalization," *BIT*, **7**, pp. 1-21 (1967).
- [Bla99] S. Blackman and R. Popoli, *Design and Analysis of Modern Tracking Systems*, Artech House, Dedham, Massachusetts, 1999.
- [Boo52] R. Booton, "An optimization theory for time-varying linear systems with non-stationary statistical inputs," *Proceedings of the IRE*, **40**, pp. 977-981 (August 1952).
- [Bor79] P. Borjesson and C. Sundberg, "Simple approximations of the error function $Q(x)$ for communications applications," *IEEE Transactions on Communications*, **COM-27**(3), pp. 639-643 (March 1979).
- [Boz94] S. Bozic, *Digital and Kalman Filtering*, Halsted Press, New York, 1994.
- [Bra89] K. Brammer and G. Siffing, *Kalman-Bucy Filters*, Artech House, Dedham, Massachusetts, 1989.
- [Bro91] P. Brockwell and R. Davis, *Time Series: Theory and Methods*, Springer-Verlag, New York, 1991.
- [Bro98] E. Brookner, *Tracking and Kalman Filtering Made Easy*, John Wiley & Sons, New York, 1998.
- [Bro96] R. Brown and P. Hwang, *Introduction to Random Signals and Applied Kalman Filtering*, John Wiley & Sons, New York, 1996.
- [Bry65] A. Bryson and D. Johanson, "Linear filtering for time-varying systems using measurements containing colored noise," *IEEE Transactions on Automatic Control*, **AC-10**(1), pp. 4-10 (January 1965).
- [Bry68] A. Bryson and L. Henrikson, "Estimation using sampled data containing sequentially correlated noise," *Journal of Spacecraft and Rockets*, **5**(6), pp. 662-665 (June 1968).
- [Buc65] R. Bucy, "Nonlinear filtering," *IEEE Transactions on Automatic Control*, **10**(1), p. 198 (January 1965).
- [Buc67] R. Bucy, "Global theory of the Riccati equation," *Journal of Computer and System Sciences*, **1**, pp. 349-361 (December 1967).
- [Buc68] R. Bucy and P. Joseph, *Filtering for Stochastic Processes with Applications to Guidance*, John Wiley & Sons, New York, 1968.
- [Buc69] R. Bucy, "Bayes theorem and digital realizations for non-linear filters," *The Journal of the Astronautical Sciences*, **17**(2), pp. 80-94 (September-October 1969).
- [Buc71] R. Bucy and K. Senne, "Digital synthesis of non-linear filters," *Automatica*, **7**(3), pp. 287-298 (May 1971).

- [Bur93] J. Burl, "A reduced order extended Kalman filter for sequential images containing a moving object," *IEEE Transactions on Image Processing*, **2**(3), pp. 285-295 (July 1993).
- [Bur99] J. Burl, *Linear Optimal Control*, Addison-Wesley, Reading, Massachusetts, 1999.
- [Cai88] P. Caines, *Linear Stochastic Systems*, John Wiley & Sons, New York, 1988.
- [Cam68] A. Cameron, "Control and estimation of linear systems with nongaussian *a priori* distributions," *Sixth Annual Allerton Conference on Circuit and System Theory* (1968).
- [Cam01] M. Campbell and S. Brunke, "Nonlinear estimation of aircraft models for on-line control customization," *IEEE Aerospace Conference*, pp. 621-628 (2001).
- [Car73] N. Carlson, "Fast triangular formulation of the square root filter," *AIAA Journal*, **11**(9), pp. 1259-1265 (September 1973).
- [Car99] J. Carpenter, P. Clifford, and P. Fearnhead, "Improved particle filter for nonlinear problems," *IEE Proceedings Part F: Radar and Sonar Navigation*, **146**, pp. 2-7 (February 1999).
- [Cat89] D. Catlin, *Estimation, Control, and the Discrete Kalman Filter*, Springer-Verlag, New York, 1989.
- [Cha47] S. Chandrasekhar, "On the radiative equilibrium of a stellar atmosphere, Part XXI," *The Astrophysical Journal*, **106**, pp. 152-216 (1947).
- [Cha48] S. Chandrasekhar, "On the radiative equilibrium of a stellar atmosphere, Part XXII," *The Astrophysical Journal*, **107**, pp. 48-72 (1948).
- [Cha96] S. Charleston and M. Azimi-Sadjadi, "Reduced order Kalman filtering for the enhancement of respiratory sounds," *IEEE Transactions on Biomedical Engineering*, **43**(4), pp. 421-424 (April 1996).
- [Che99] C. Chen, *Linear System Theory and Design*, Oxford University Press, 1999.
- [Che93] G. Chen, *Approximate Kalman Filtering*, World Scientific, Singapore, 1993.
- [Che85] H. Chen, *Recursive Estimation and Control for Stochastic Systems*, John Wiley & Sons, New York, 1985.
- [Chi85] T. Chia, *Parameter Identification and State Estimation of Constrained Systems*, Doctoral Dissertation, Case Western Reserve University, Cleveland, Ohio, 1985.
- [Chu87] C. Chui and G. Chen, *Kalman Filtering with Real-Time Applications*, Springer-Verlag, New York, 1987.
- [Coh96] S. Cohn and R. Todling, "Approximate data assimilation scheme for stable and unstable dynamics," *Journal of the Meteorological Society of Japan*, **74**(1), pp. 63-75 (1996).
- [Cox64] H. Cox, "On the estimation of state variables and parameters for noisy dynamic systems," *IEEE Transactions on Automatic Control*, **AC-9**(1), pp. 5-12 (January 1964).
- [Cox73] H. Coxeter, *Regular Polytopes*, Dover, New York, 1973.
- [Cra04] J. Crassidis and J. Junkins, *Optimal Estimation of Dynamic Systems*, CRC Press, New York, 2004.
- [Cri02] D. Crisan and A. Doucet, "A survey of convergence results on particle filtering methods for practitioners," *IEEE Transactions on Signal Processing*, **50**(3), pp. 736-746 (March 2002).

- [Dar84] J. Darragh and D. Looze, "Noncausal minimax linear state estimation for systems with uncertain second-order statistics," *IEEE Transactions on Automatic Control*, **AC-29**(6), pp. 555-557 (June 1984).
- [Dav77] M. Davis, *Linear Estimation and Stochastic Control*, Chapman & Hall, London, 1977.
- [deF71] R. de Figueiredo and Y. Jan, "Spline filters," *Symposium on Nonlinear Estimation Theory and Its Applications*, pp. 127-138 (1971).
- [deF74] B. de Finetti, *Theory of Probability: A Critical Introductory Treatment*, John Wiley & Sons, New York, 1974.
- [deF02] N. de Freitas, "Rao-Blackwellised particle filtering for fault diagnosis," *IEEE Aerospace Conference*, pp. 1767-1772 (2002).
- [Deu65] R. Deutsch, *Estimation Theory*, Prentice-Hall, Englewood Cliffs, New Jersey, 1965.
- [Dev01] L. Devroye and G. Lugosi, *Combinatorial Methods in Density Estimation*, Springer-Verlag, New York, 2001.
- [Dew03] P. Dewallef and O. Leonard, "On-line performance monitoring and engine diagnostic using robust Kalman filtering techniques," *ASME Turbo Expo*, Paper GT2003-38379 (2003).
- [Dew04] P. Dewallef, O. Leonard, and K. Mathioudakis, "On-line aircraft engine diagnostic using a soft-constrained Kalman filter," *ASME Turbo Expo*, Paper GT2004-53539 (2004).
- [Dez92] F. Deza, E. Busvelle, J. Gauthier, and D. Rakotopara, "High gain estimation for nonlinear systems," *Systems and Control Letters*, **18**, pp. 295-299 (1992).
- [Did85] G. Diderrich, "The Kalman filter from the perspective of the Goldberger-Thiel estimators," *The American Statistician*, **39**(3), pp. 193-198 (August 1985).
- [Did95] G. Didinsky, Z. Pan, and T. Basar, "Parameter identification for uncertain plants using H^∞ methods," *Automatica*, **31**(9), pp. 1227-1250 (September 1995).
- [Dou01] A. Doucet, N. de Freitas, and N. Gordon (eds.), *Sequential Monte Carlo Methods in Practice*, Springer-Verlag, New York, 2001.
- [Dun44] W. Duncan, "Some devices for the solution of large sets of simultaneous linear equations (with an appendix on the reciprocation of partitioned matrices)," *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science, Seventh Series*, **35**, pp. 660-670 (1944).
- [Dye69] P. Dyer and S. McReynolds, "Extension of square-root filtering to include process noise," *Journal of Optimization Theory and Applications*, **3**(6), pp. 444-459 (1969).
- [Fad59] V. Fadeeva, *Computational Methods of Linear Algebra*, Dover, New York, 1959.
- [Fey63] R. Feynman, R. Leighton, and M. Sands, *The Feynman Lectures on Physics: Volume 1*, Addison-Wesley, Reading, Massachusetts, 1963.
- [Fit71] R. Fitzgerald, "Divergence of the Kalman filter," *IEEE Transactions on Automatic Control*, **AC-16**, pp. 736-747 (December 1971).
- [Fle81] R. Fletcher, *Practical Methods of Optimization - Volume 2: Constrained Optimization*, John Wiley & Sons, New York, 1981.
- [Fom00] S. Fomin and I. Gelfand, *Calculus of Variations*, Dover, New York, 2000.

- [Fra69] D. Fraser and J. Potter, "The optimum linear smoother as a combination of two optimum linear filters," *IEEE Transactions on Automatic Control*, **14**(4), pp. 387-390 (August 1969).
- [Fri66] B. Friedland and I. Bernstein, "Estimation of the state of a nonlinear process in the presence of nongaussian noise and disturbances," *Journal of the Franklin Institute*, **281**(6), pp. 455-480 (June 1966).
- [Fri89] B. Friedland, "On the properties of reduced-order Kalman filters," *IEEE Transactions on Automatic Control*, **34**(3), pp. 321-324 (March 1989).
- [Gau04] C. Gauss, *Theory of Motion of the Heavenly Bodies Moving About the Sun in Conic Sections: A Translation of Theoria Motus*, Dover, New York, 2004.
- [Gee97] J. De Geeter, H. Van Brussel, and J. De Schutter, "A smoothly constrained Kalman filter," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, **19**(10), pp. 1171-1177 (October 1997).
- [Gel74] A. Gelb (ed.), *Applied Optimal Estimation*, MIT Press, Cambridge, Massachusetts, 1974.
- [Gil01] W. Gilks and C. Berzuini, "Following a moving target – Monte Carlo inference for dynamic Bayesian models," *Journal of the Royal Statistical Society, B*, **63**, pp. 127-146 (2001).
- [Gil81] P. Gill, W. Murray, and M. Wright, *Practical Optimization*, Academic Press, New York, 1981.
- [Gli94] L. Glielmo, P. Marino, R. Setola, and F. Vasca, "Reduced Kalman filtering for indirect adaptive control of the induction motor," *International Journal of Adaptive Control and Signal Processing*, **8**, pp. 527-541 (1994).
- [Gol65] G. Golub, "Numerical methods for solving linear least squares problems," *Numerische Mathematik*, **7**, pp. 206-216 (1965).
- [Gol89] G. Golub and C. Van Loan, *Matrix Computations*, The Johns Hopkins University Press, Baltimore, Maryland, 1989.
- [Goo84] G. Goodwin and K. Sin, *Adaptive Filtering, Prediction and Control*, Prentice-Hall, Englewood Cliffs, New Jersey, 1984.
- [Goo05a] G. Goodwin, M. Seron, and J. De Dona, *Constrained control and estimation*, Springer-Verlag, London, 2005.
- [Goo05b] G. Goodwin, J. De Dona, M. Seron, and X. Zhuo, "Lagrangian duality between constrained estimation and control," *Automatica*, **41**(6), pp. 935-944 (June 2005).
- [Gor93] N. Gordon, D. Salmond, and A. Smith, "Novel approach to nonlinear/non-Gaussian Bayesian state estimation," *IEEE Proceedings-F*, **140**(2), pp. 107-113 (April 1993).
- [Gra93] J. Gray and W. Murray, "A derivation of an analytic expression for the tracking index for the alpha-beta-gamma filter," *IEEE Transactions on Aerospace and Electronic Systems*, **29**(3), pp. 1064-1065 (July 1993).
- [Gre01] M. Grewal and A. Andrews, *Kalman Filtering*, John Wiley & Sons, New York, 2001.
- [Gri97] K. Grigoriadis and J. Watson, "Reduced-order H_∞ and $L_2 - L_\infty$ filtering via linear matrix inequalities," *IEEE Transactions on Aerospace and Electronic Systems*, **33**(4), pp. 1326-1338 (October 1997).
- [Gri88] M. Grumble, " H_∞ design of optimal linear filters," in *Linear Circuits, Systems and Signal Processing: Theory and Applications*, C. Byrnes, C. Martin, and R. Sacks (Eds.) North-Holland, Amsterdam, The Netherlands, 1988, pp. 533-540.

- [Gri90] M. Grimble and A. El Sayed, "Solution of the H_∞ optimal linear filtering problem for discrete-time systems," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, **38**(7), pp. 1092-1104 (July 1990).
- [Gri91a] M. Grimble, " H_∞ fixed lag smoothing filter for scalar systems," *IEEE Transactions on Signal Processing*, **39**(9), pp. 1955-1963 (September 1991).
- [Gri91b] M. Grimble and M. Johnson, " H_∞ robust control design – A tutorial review," *Computing and Control Engineering Journal*, **2**(6), pp. 275-282 (November 1991).
- [Hab03] S. Habibi and R. Burton, "The variable structure filter," *ASME Journal of Dynamic Systems, Measurement, and Control*, **125**(3), pp. 287-293 (September 2003).
- [Had91] W. Haddad, D. Bernstein, and D. Mustafa, "Mixed-norm H_2/H_∞ regulation and estimation: The discrete-time case," *Systems and Control Letters*, **16**, pp. 235-247 (1991).
- [Hae98] D. Haessig and B. Friedland, "Separate-bias estimation with reduced-order Kalman filters," *IEEE Transactions on Automatic Control*, **43**(7), pp. 983-987. (July 1998).
- [Hal81] A. Hald, "T. N. Thiele's contributions to statistics," *International Statistics Review*, **49**, pp. 1-20 (1981).
- [Har89] A. Harvey, *Forecasting, Structural Time Series Models and the Kalman Filter*, Cambridge University Press, New York, 1989.
- [Has95] B. Hassibi and T. Kailath, " H^∞ adaptive filtering," *IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 949-952 (1995).
- [Has96a] B. Hassibi, A. Sayed, and T. Kailath, "Linear estimation in Krein spaces – Part I: Theory," *IEEE Transactions on Automatic Control*, **41**(1), pp. 18-33 (January 1996).
- [Has96b] B. Hassibi, A. Sayed, and T. Kailath, "Linear estimation in Krein spaces – Part II: Applications," *IEEE Transactions on Automatic Control*, **41**(1), pp. 34-49 (January 1996).
- [Has99] B. Hassibi, A. Sayed, T. Kailath, *Indefinite-Quadratic Estimation and Control*, SIAM, Philadelphia, Pennsylvania, 1999.
- [Hay98] S. Hayward, "Constrained Kalman filter for least-squares estimation of time-varying beamforming weights," in *Mathematics in Signal Processing IV*, J. McWhirter and I. Proudler (Eds.), Oxford University Press, pp. 113-125, 1998.
- [Hec71] C. Hecht, "Digital realization of non-linear filters," *Symposium on Nonlinear Estimation Theory and Its Applications*, pp. 152-158 (1971).
- [Hec73] C. Hecht, "Systems identification using Bayesian estimation," *Symposium on Nonlinear Estimation Theory and Its Applications*, pp. 107-113 (1973).
- [Hef66] H. Hefes, "The effect of erroneous models on the Kalman filter response," *IEEE Transactions on Automatic Control*, **AC-11**, pp. 541-543 (July 1966).
- [Hel94] Y. Hel-Or and M. Werman, "Recognition and localization of articulated objects," *IEEE Workshop on Motion of Non-Rigid and Articulated Objects*, pp. 116-123 (November 1994).
- [Hen81] H. Henderson and S. Searle, "On deriving the inverse of a sum of matrices," *SIAM Review*, **23**(1), pp. 53-60 (January 1981).
- [Hen82] R. Henriksen, "The truncated second-order nonlinear filter revisited," *IEEE Transactions on Automatic Control*, **AC-27**(1), pp. 247-251 (February 1982).

- [Hid03] C. Hide, T. Moore, and M. Smith, "Adaptive Kalman filtering for low-cost INS/GPS," *The Journal of Navigation*, **56**, pp. 143-152 (2003).
- [Ho64] Y. Ho and R. Lee, "A Bayesian approach to problems in stochastic estimation and control," *IEEE Transactions on Automatic Control*, **AC-9**, pp. 333-339 (1964).
- [Ho02] Y. Ho and D. Pepyne, "Simple explanation of the no-free-lunch theorem and its implications," *Journal of Optimization Theory and Applications*, **155**(3), 549-570. (December 2002).
- [Hor85] R. Horn and C. Johnson, *Matrix Analysis*, Cambridge University Press, New York, 1985.
- [Hou53] A. Householder, *Principles of Numerical Analysis*, McGraw-Hill, New York, 1953.
- [Hou64] A. Householder, *The Theory of Matrices in Numerical Analysis*, Blaisdell, New York, 1964.
- [Hsi96] F. Hsiao and S. Pan, "Robust Kalman filter synthesis for uncertain multiple time-delay stochastic systems," *Journal of Dynamic Systems, Measurement, and Control*, **118**, pp. 803-808 (December 1996).
- [Hug88] R. Hughes, *Analog Automatic Control Loops in Radar and EW*, Artech House, Dedham, Massachusetts, 1988.
- [Hun03] Y. Hung and F. Yang, "Robust H_∞ filtering with error variance constraints for discrete time-varying systems with uncertainty," *Automatica*, **39**(7), pp. 1185-1194 (July 2003).
- [Hut84] C. Hutchinson, "The Kalman filter applied to aerospace and electronic systems," *IEEE Transactions on Aerospace and Electronic Systems*, **AES-20**(4), pp. 500-504 (July 1984).
- [Iba01] Y. Iba, "Population Monte Carlo algorithms," *Transactions of the Japanese Society for Artificial Intelligence*, **16**(2), pp. 279-286. (2001).
- [Ion99] V. Ionescu, C. Oara, and M. Weiss, *Generalized Riccati Theory and Robust Control*, John Wiley & Sons, New York, 1999.
- [Isa96] M. Isard and A. Blake, "Contour tracking by stochastic propagation of conditional density," *European Conference on Computer Vision*, Cambridge, pp. 343-356, 1996.
- [Isa98a] M. Isard and A. Blake, "Condensation – Conditional density propagation for visual tracking," *International Journal of Computer Vision*, **29**(1), pp. 5-28. (1998).
- [Isa98b] M. Isard and A. Blake, "ICondensation: Unifying low-level and high-level tracking in a stochastic framework," *Lecture Notes in Computer Science*, **1406**, pp. 893-908. (1998).
- [Jaz66] A. Jazwinski, "Filtering for nonlinear dynamical systems," *IEEE Transactions on Automatic Control*, **AC-11**(5), pp. 765-766 (October 1966).
- [Jaz69] A. Jazwinski, "Adaptive filtering," *Automatica*, **5**(4), pp. 475-485 (July 1969).
- [Jaz70] A. Jazwinski, *Stochastic Processes and Filtering Theory*, Academic Press, New York, 1970.
- [Joh69] G. Johnson, "A deterministic theory of estimation and control," *IEEE Transactions on Automatic Control*, **14**(4), pp. 380-384 (August 1969).
- [Jor68] T. Jordan, "Experiments on error growth associated with some linear least-squares procedures," *Mathematics of Computation*, **22**, pp. 579-588 (1968).

- [Jua93] J. Juang, *Applied System Identification*, Prentice-Hall, Upper Saddle River, New Jersey, 1993.
- [Jul95] S. Julier, J. Uhlmann, and H. Durrant-Whyte, "A new approach for filtering nonlinear systems," *American Control Conference*, pp. 1628-1632 (1995).
- [Jul00] S. Julier, J. Uhlmann, and H. Durrant-Whyte, "A new method for the nonlinear transformation of means and covariances in filters and estimators," *IEEE Transactions on Automatic Control*, **45**(3), pp. 477-482 (March 2000).
- [Jul02a] S. Julier and J. Uhlmann, "Reduced sigma point filters for the propagation of means and covariances through nonlinear transformations," *American Control Conference*, pp. 887-892, 2002.
- [Jul03] S. Julier, "The spherical simplex unscented transformation," *American Control Conference*, pp. 2430-2434, 2003.
- [Jul04] S. Julier and J. Uhlmann, "Unscented filtering and nonlinear estimation," *Proceedings of the IEEE*, **92**(3), pp. 401-422 (March 2004).
- [Kai68] T. Kailath, "An innovations approach to least-squares estimation – Part I: Linear filtering in additive white noise," *IEEE Transactions on Automatic Control*, **AC-13**(6), pp. 646-655 (December 1968).
- [Kai73] T. Kailath, "Some new algorithms for recursive estimation in constant linear systems," *IEEE Transactions on Information Theory*, **IT-19**(6), pp. 750-760 (November 1973).
- [Kai74] T. Kailath, "A view of three decades of linear filtering theory," *IEEE Transactions on Information Theory*, **IT-20**(2), pp. 146-181 (March 1974).
- [Kai77] T. Kailath (ed.), *Linear Least Squares Estimation*, Dowden, Hutchinson & Ross, Stroudsburg, Pennsylvania, 1977.
- [Kai80] T. Kailath, *Linear Systems*, Prentice-Hall, Englewood Cliffs, New Jersey, 1980.
- [Kai81] T. Kailath, *Lectures on Wiener and Kalman Filtering*, Springer-Verlag, New York, 1981.
- [Kai00] T. Kailath, A. Sayed, and B. Hassibi, *Linear Estimation*, Prentice-Hall, Upper Saddle River, New Jersey, 2000.
- [Kal84] P. Kalata, "The tracking index: A generalized parameter for α - β and α - β - γ target trackers," *IEEE Transactions on Aerospace and Electronic Systems*, **AES-20**(2), pp. 174-182.
- [Kal60] R. Kalman, "A new approach to linear filtering and prediction problems," *ASME Journal of Basic Engineering*, **82**, pp. 35-45 (March 1960).
- [Kal60b] R. Kalman, "Contributions to the theory of optimal control," *Boletin de la Sociedad Matematica Mexicana*, **5**, pp. 102-119 (1960).
- [Kal60c] R. Kalman, "On the general theory of control systems," *Proceedings of the 1st IFAC Congress*, pp. 481-492, Butterworth's, London, 1960.
- [Kal61] R. Kalman and R. Bucy, "New results in linear filtering and prediction theory," *ASME Journal of Basic Engineering*, **83**, pp. 95-108 (March 1961).
- [Kal63] R. Kalman, Y. Ho, and K. Narendra, "Controllability of linear dynamical systems," *Contributions to Differential Equations*, **1**(2), pp. 189-213 (1963).
- [Kam71] P. Kaminski, A. Bryson, and S. Schmidt, "Discrete square root filtering: A survey of current techniques," *IEEE Transactions on Automatic Control*, **AC-16**, pp. 727-736 (December 1971).

- [Kan95] K. Kanazawa, D. Koller, and S. Russell, "Stochastic simulation algorithms for dynamics probabilistic networks," *Eleventh Annual Conference on Uncertainty in AI*, pp. 346-351, 1995.
- [Kas96] E. Kaskurewicz and A. Bhaya, "Discrete-time state estimation with two counters and measurement delay," *IEEE Conference on Decision and Control*, pp. 1472-1476, 1996.
- [Kel98] J. Keller and M. Darouach, "Reduced-order Kalman filter with unknown inputs," *Automatica*, **34**(11), pp. 1463-1468 (November 1998).
- [Kel99] J. Keller, "A new strategy for designing a reduced-order Kalman filter," *International Journal of Systems Science*, **30**(11), pp. 1161-1166 (1999).
- [Ken61] M. Kendall, "Daniel Bernoulli on maximum likelihood," *Biometrika*, **48**, pp. 1-18 (1961).
- [Kit87] G. Kitagawa, "Non-Gaussian state-space modelling of non-stationary time series (with discussion)," *Journal of the American Statistical Association*, **82**(400), pp. 1032-1063 (December 1987).
- [Kit89] G. Kitagawa, "Non-Gaussian seasonal adjustment," *Computers and Mathematics with Applications*, **18**(6,7), pp. 503-514 (1989).
- [Kit94] G. Kitagawa, "The two-filter formula for smoothing and an implementation of the Gaussian-sum smoother," *Annals of the Institute of Statistical Mathematics*, **46**(4), pp. 605-623 (1994).
- [Kit96a] G. Kitagawa, *Smoothness Priors Analysis of Time Series*, Springer-Verlag, New York, 1996.
- [Kit96] G. Kitagawa, "Monte Carlo filter and smoother for non-Gaussian nonlinear state space models," *Journal of Computational and Graphical Statistics*, **5**(1), pp. 1-25 (1996).
- [Ko06] S. Ko and R. Bitmead, "State estimation for linear systems with state equality constraints," *Automatica*, 2006.
- [Kob03] T. Kobayashi and D.L. Simon, "Application of a bank of Kalman filters for aircraft engine fault diagnosis," *ASME Turbo Expo*, Atlanta, paper GT2003-38550, June 2003.
- [Kol41] A. Kolmogorov, "Interpolation and extrapolation of stationary random sequences," in *Selected Works of A. N. Kolmogorov, Volume II*, V. Tikhomirov (Ed.), Kluwer Academic Publishers, Dordrecht, The Netherlands, pp. 272-280 (1992).
- [Kop63] R. Kopp and R. Orford, "Linear regression applied to system identification for adaptive control systems," *AIAA Journal*, **1**, pp. 2300-2306 (October 1963).
- [Kos04] S. Kosanam and D. Simon, "Kalman filtering with uncertain noise covariances," *Intelligent Systems and Control*, Honolulu, Hawaii, pp. 375-379, August 2004.
- [Kra88] S. Kramer and H. Sorenson, "Recursive Bayesian estimation using piece-wise constant approximations," *Automatica*, **24**(6), pp. 789-801 (November 1988).
- [Kri84] V. Krishnan, *Nonlinear Filtering and Smoothing: An Introduction to Martingales, Stochastic Integrals and Estimation*, John Wiley & Sons, New York, 1984.
- [Kus67a] H. Kushner, "Dynamical equations for optimal nonlinear filtering," *Journal of Differential Equations*, **3**, pp. 179-190 (1967).
- [Kus67] H. Kushner, "Approximations to optimal nonlinear filters," *IEEE Transactions on Automatic Control*, **AC-12**(5), pp. 546-556 (October 1967).

- [Kwa72] H. Kwakernaak and R. Sivan, *Linear Optimal Control Systems*, John Wiley & Sons, New York, 1972.
- [Kwo04] C. Kwok, D. Fox, and M. Meila, "Real-time particle filters," *Proceedings of the IEEE*, **92**(3), pp. 469-484 (March 2004).
- [Kwo03] S. Kwon and W. Chung, "Combined synthesis of state estimator and perturbation observer," *ASME Journal of Dynamic Systems, Measurement, and Control*, **125**(1), pp. 19-26 (March 2003).
- [Lai74] D. Lainiotis, "Estimation: A brief survey," *Information Sciences*, **7**, pp. 191-202 (1974).
- [Lan95] P. Lancaster and L. Rodman, *Algebraic Riccati Equations*, Oxford University Press, New York, 1995.
- [Lar98] T. Larsen, N. Andersen, O. Ravn, and N. Poulsen, "Incorporation of time delayed measurements in a discrete-time Kalman filter," *IEEE Conference on Decision and Control*, pp. 3972-3977, 1998.
- [Lau81] S. Lauritzen, "Time series analysis in 1880: A discussion of contributions made by T. N. Thiele," *International Statistics Review*, **49**, pp. 319-331 (1981).
- [Lee64] R. Lee, *Optimal Estimation, Identification, and Control*, The MIT Press, Cambridge, Massachusetts, 1964.
- [Lew51] K. Lewin, *Field Theory in Social Science; Selected Theoretical Papers*, Harper & Row, New York, 1951.
- [Lew86a] F. Lewis, *Optimal Control*, John Wiley & Sons, New York, 1986.
- [Lew86b] F. Lewis, *Optimal Estimation*, John Wiley & Sons, New York, 1986.
- [Lew97] H. Lewis, *Foundations of Fuzzy Control*, Plenum, New York, 1997.
- [Lie67] P. Liebelt, *An Introduction to Optimal Estimation*, Addison-Wesley, Reading, Massachusetts, 1967.
- [Liu80] C. Liu and S. Marcus, "Estimator performance for a class of nonlinear estimation problems," *IEEE Transactions on Automatic Control*, **AC-25**(2), pp. 299-302 (April 1980).
- [Liu98] J. Liu and R. Chen, "Sequential Monte Carlo methods for dynamical systems," *Journal of the American Statistical Association*, **93**, pp. 1032-1044 (1998).
- [Lju98] L. Ljung, *System Identification: Theory for the User*, Prentice-Hall, Upper Saddle River, New Jersey, 1998.
- [Lue84] D. Luenberger, *Linear and Nonlinear Programming*, Addison-Wesley, Reading, Massachusetts, 1984.
- [Mac99] J. McCormick and A. Blake, "A probabilistic exclusion principle for tracking multiple objects," *International Conference on Computer Vision*, pp. 572-578, 1999.
- [Mah04a] K. Mahata and T. Soderstrom, "Improved estimation performance using known linear constraints," *Automatica*, **40**(8), pp. 1307-1318 (August 2004).
- [Mah04b] M. Mahmoud, "Resilient linear filtering of uncertain systems," *Automatica*, **40**(10), pp. 1797-1802 (October 2004).
- [Mai84] J. Maindonald, *Statistical Computation*, John Wiley & Sons, New York, 1984.
- [Mal01] M. Mallick, S. Coraluppi, and C. Carthel, "Advances in asynchronous and decentralized estimation," *IEEE Aerospace Conference*, pp. 1873-1888, 2001.
- [Mar05] H. Marquez and M. Riaz, "Robust state observer design with application to an industrial boiler system," *Control Engineering Practice*, **13**(6), pp. 713-728 (June 2005).

- [Mas75] C. Masreliez, "Approximate non-Gaussian filtering with linear state and observation relations," *IEEE Transactions on Automatic Control*, **AC-20**, pp. 107-110 (February 1975).
- [Mas77] C. Masreliez and R. Martin, "Robust Bayesian estimation for the linear model and robustifying the Kalman filter," *IEEE Transactions on Automatic Control*, **AC-22**(3), pp. 361-371 (June 1977).
- [May79] P. Maybeck, *Stochastic Models, Estimation, and Control - Volume 1*, Academic Press, New York, 1979.
- [May82] P. Maybeck, *Stochastic Models, Estimation, and Control - Volume 2*, Academic Press, New York, 1982.
- [May84] P. Maybeck, *Stochastic Models, Estimation, and Control - Volume 3*, Academic Press, New York, 1984.
- [McG74] T. McGarty, *Stochastic Systems and State Estimation*, John Wiley & Sons, New York, 1974.
- [McG85] L. McGee and S. Schmidt, "Discovery of the Kalman filter as a practical tool for aerospace and industry," NASA Technical Memo 86847, November 1985.
- [Mc75] S. McReynolds, "Multidimensional Hermite-Gaussian quadrature formulae and their application to nonlinear estimation," *Symposium on Nonlinear Estimation Theory and Its Applications*, pp. 188-191 (1975).
- [Mec49] E. Mechler, J. Russell, and M. Preston, "The basis for the optimum aided-tracking time constant," *Journal of the Franklin Institute*, **248**, pp. 327-334 (October 1949).
- [Med69] J. Meditch, *Stochastic Optimal Linear Estimation and Control*, McGraw-Hill, New York, 1969.
- [Med73] J. Meditch, "A survey of data smoothing for linear and nonlinear dynamic systems," *Automatica*, **9**(2), pp. 151-162 (March 1973).
- [Meh70] R. Mehra, "On the identification of variances and adaptive Kalman filtering," *IEEE Transactions on Automatic Control*, **AC-15**(2), pp. 175-184 (April 1970).
- [Meh72] R. Mehra, "Approaches to adaptive filtering," *IEEE Transactions on Automatic Control*, **AC-17**(5), pp. 693-698 (October 1972).
- [Men87] J. Mendel, *Lessons in Digital Estimation Theory*, Prentice-Hall, Englewood Cliffs, New Jersey, 1987.
- [Met49] N. Metropolis and S. Ulam, "The Monte Carlo method," *Journal of the American Statistical Association*, **44**(247), pp. 335-341 (September 1949).
- [Min93] G. Minkler and J. Minkler, *Theory and Application of Kalman Filtering*, Magellan Book Company, Palm Bay, Florida, 1993.
- [Mir90] L. Mirsky, *An Introduction to Linear Algebra*, Dover, New York, 1990.
- [Mol03] C. Moler and C. Van Loan, "Nineteen dubious ways to compute the exponential of a matrix, twenty-five years later," *SIAM Review*, **45**(1), pp. 3-49 (January 2003).
- [Moo00] T. Moon and W. Stirling, *Mathematical Methods and Algorithms for Signal Processing*, Prentice-Hall, Upper Saddle River, New Jersey, 2000.
- [Mor98] P. Moral, "Measure valued processes and interacting particle systems: Application to non-linear filtering problems," *Annals of Applied Probability*, **8**(2), pp. 438-495, 1998.

- [Mor03] R. Morales-Menendez, N. de Freitas, and D. Poole, "Estimation and control of industrial processes with particle filters," *American Control Conference*, pp. 579-584, 2003.
- [Mor78] M. Morf, R. Levy, and T. Kailath, "Square-root algorithms for the continuous-time linear least-square estimation problem," *IEEE Transactions on Automatic Control*, **AC-23**(5), pp. 907-911 (October 1978).
- [Mul91] P. Muller, "Monte Carlo integration in general dynamic models," *Contemporary Mathematics*, **115**, pp. 145-163 (1991).
- [Mye76] K. Myers and B. Tapley, "Adaptive sequential estimation with unknown noise statistics," *IEEE Transactions on Automatic Control*, **AC-21**(4), pp. 520-523 (August 1976).
- [Nag87] K. Nagpal, R. Helmick, and C. Sims, "Reduced-order estimation: Part I. Filtering," *International Journal of Control*, **45**, pp. 1867-1888 (1987).
- [Nag91] K. Nagpal and P. Khargonekar, "Filtering and smoothing in an H^∞ setting," *IEEE Transactions on Automatic Control*, **36**(2), pp. 152-166 (February 1991).
- [Nah69] N. Nahi, "Optimal recursive estimation with uncertain observation," *IEEE Transactions on Information Theory*, **IT-15**(4), pp. 457-462 (July 1969).
- [Nel87] E. Nelson, *Radically Elementary Probability Theory*, Princeton University Press, Princeton, New Jersey, 1987.
- [Net78] M. Netto, L. Gimeno, and M. Mendes, "On the optimal and suboptimal nonlinear filtering problem for discrete-time systems," *IEEE Transactions on Automatic Control*, **AC-23**(6), pp. 1062-1067 (December 1978).
- [Nis66] T. Nishimura, "On the a priori information in sequential estimation problems," *IEEE Transactions on Automatic Control*, **AC-11**, pp. 197-204 (April 1966).
- [Nor00] M. Norgaard, N. Poulsen, and O. Ravn, "New developments in state estimation for nonlinear systems," *Automatica*, **36**(11), pp. 1627-1638 (November 2000).
- [OC096] J. O'Connor and E. Robertson, *Matrices and Determinants*, http://www-history.mcs.st-andrews.ac.uk/HistTopics/Matrices_and_determinants.html (1996).
- [Ott85] P. Otter, *Dynamic Feature Space Modelling, Filtering and Self-Tuning Control of Stochastic Systems*, Springer-Verlag, New York, 1985.
- [Pan96] Z. Pan and T. Basar, "Parameter identification for uncertain linear systems with partial state measurements under an H^∞ criterion," *IEEE Transactions on Automatic Control*, **41**(9), pp. 1295-1311 (September 1996).
- [Pap02] A. Papoulis and S. Pillai, *Probability, Random Variables and Stochastic Processes*, McGraw-Hill, New York, 2002.
- [Pat98] A. Patti, A. Tekalp, and M. Sezan, "A new motion-compensated reduced-order model Kalman filter for space-varying restoration of progressive and interlaced video," *IEEE Transactions on Image Processing*, **7**(4), pp. 543-554 (April 1998).
- [Pee01] P. Peebles, *Probability, Random Variables, and Random Signal Principles*, McGraw-Hill, New York, 2001.
- [Pha98] D. Pham, J. Verron, and M. Roubaud, "A singular evolutive extended Kalman filter for data assimilation in oceanography," *Journal of Marine Systems*, **16**, pp. 323-240 (1998).
- [Pit99] M. Pitt and N. Shephard, "Filtering via simulation: Auxiliary particle filters," *Journal of the American Statistical Association*, **94**(446), pp. 590-599 (1999).

- [Pla50] R. Plackett, "Some theorems in least squares," *Biometrika*, **37**, pp. 149-157 (1950).
- [Poo81] V. Poor and D. Looze, "Minimax state estimation for linear stochastic systems with noise uncertainty," *IEEE Transactions on Automatic Control*, **AC-26**(4), pp. 902-906 (August 1981).
- [Por88] J. Porrill, "Optimal combination and constraints for geometrical sensor data," *International Journal of Robotics Research*, **7**(6), pp. 66-77 (December 1988).
- [Pou86] M. Poubelle, I. Petersen, M. Gevers, and R. Bitmead, "A miscellany of results on an equation of Count J. F. Riccati," *IEEE Transactions on Automatic Control*, **AC-31**(7), pp. 651-654 (July 1986).
- [Pre92] W. Press, B. Flannery, S. Teukolsky, and W. Vetterling, *Numerical Recipes in C: The Art of Scientific Computing*, Cambridge University Press, New York, 1992.
- [Rao03] C. Rao, J. Rawlings, and D. Mayne, "Constrained state estimation for non-linear discrete-time systems: Stability and moving horizon approximations," *IEEE Transactions on Automatic Control*, **48**(2), pp. 246-258 (February 2003).
- [Rau65] H. Rauch, F. Tung, and C. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, **3**(8), pp. 1445-1450 (August 1965).
- [Rei72] W. Reid, *Riccati Differential Equations*, Academic Press, New York, 1972.
- [Rei98] K. Reif, F. Sonnemann, and R. Unbehauen, "An EKF-based nonlinear observer with a prescribed degree of stability," *Automatica*, **34**(9), pp. 1119-1123 (September 1998).
- [Rei99] K. Reif, F. Sonnemann, and R. Unbehauen, "Nonlinear state observation using H_∞ -filtering Riccati design," *IEEE Transactions on Automatic Control*, **44**(1), pp. 203-208 (January 1999).
- [Rhe89] I. Rhee and J. Speyer, "A game theoretic controller and its relationship to H_∞ and linear-exponential-Gaussian synthesis," *IEEE Conference on Decision and Control*, pp. 909-915 (1989).
- [Rho71] I. Rhodes, "A tutorial introduction to estimation and filtering," *IEEE Transactions on Automatic Control*, **AC-16**(6), pp. 688-706 (December 1971).
- [Rip87] B. Ripley, *Stochastic Simulation*, John Wiley & Sons, New York, 1987.
- [Ris04] B. Ristic, S. Arulampalam, and N. Gordon, *Beyond the Kalman Filter: Particle Filters for Tracking Applications*, Artech House, Norwell, Massachusetts, 2004.
- [Rob99] C. Robert and G. Casella, *Monte Carlo Statistical Methods*, Springer-Verlag, New York, 1999.
- [Ruy85] P. Ruymgaart and T. Soong, *Mathematics of Kalman-Bucy Filtering*, Springer-Verlag, New York, 1985.
- [Saf78] M. Safonov and M. Athans, "Robustness and computational aspects of non-linear stochastic estimators and regulators," *IEEE Transactions on Automatic Control*, **AC-23**(4), pp. 717-725 (August 1978).
- [Sag71] A. Sage and J. Melsa, *Estimation Theory with Applications to Communications and Control*, McGraw-Hill, New York, 1971.
- [Sch67] F. Schlee, C. Standish, and N. Toda, "Divergence in the Kalman filter," *AIAA Journal*, **5**, pp. 1114-1120 (June 1967).
- [Sch66] S. Schmidt, "Applications of state space methods to navigation problems," in C. Leondes (Ed.), *Advances in Control Systems*, Vol. 3, pp. 293-340, Academic Press, New York, 1966.

- [Sch81] S. Schmidt, "The Kalman filter: Its recognition and development for aerospace applications," *Journal of Guidance and Control*, **4**(1), pp. 4-7 (January-February 1981).
- [Sch68] L. Schwartz and E. Stear, "A computational comparison of several nonlinear filters," *IEEE Transactions on Automatic Control*, **13**(1), pp. 83-86 (February 1968).
- [Sch73] F. Schweppe, *Uncertain Dynamic Systems*, Prentice-Hall, Englewood Cliffs, New Jersey, 1973.
- [Sea67] H. Seal, "The historical development of the Gauss linear model," *Biometrika*, **54**(1-2), pp. 1-24 (1967).
- [Sch17] I. Schur, "Über potenzreihen, die in innern des einheitskreises beschränkt sind," *Journal fuer die Reine und Angewandte Mathematik*, **147**, pp. 205-232 (1917).
- [Sch86] I. Schur, "On power series which are bounded in the interior of the unit circle, Part I," in I. Gohberg (Ed.), *I. Schur Methods in Operator Theory and Signal Processing*, pp. 31-88, Birkhauser, Boston, Massachusetts, 1986.
- [Ser81] L. Servi and Y. Ho, "Recursive estimation in the presence of uniformly distributed measurement noise," *IEEE Transactions on Automatic Control*, **26**(2), pp. 563-564 (April 1981).
- [Sha82] W. Shanahan, "Circuit models for prediction, Wiener filtering, Levinson and Kalman filters for ARMA time series," *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 266-269 (May 1982).
- [Sha92] U. Shaked and Y. Theodor, " H_∞ -optimal estimation: a tutorial," *IEEE Conference on Decision and Control*, pp. 2278-2286 (December 1992).
- [She95] X. Shen and L. Deng, "Discrete H_∞ filter design with application to speech enhancement," *IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 1504-1507 (1995).
- [She97] X. Shen and L. Deng, "Game theory approach to discrete H_∞ filter design," *IEEE Transactions on Signal Processing*, pp. 1092-1095 (April 1997).
- [She99] K. Shen, J. Crossley, A. Lun, S. Kangshen, and H. Liu, *The Nine Chapters on the Mathematical Art: Companion and Commentary*, Oxford University Press, Oxford (1999).
- [Shi98] N. Shimada, Y. Shirai, Y. Kuno, and J. Miura, "Hand gesture estimation and model refinement using monocular camera – Ambiguity limitation by inequality constraints," *Third IEEE International Conference on Automatic Face and Gesture Recognition*, pp. 268-273 (April 1998).
- [Sil86] B. Silverman, *Density Estimation for Statistics and Data Analysis*, Chapman & Hall, London, 1986.
- [Sim96] D. Simon and H. El-Sherief, "Hybrid Kalman/minimax filtering in phase-locked loops," *Control Engineering Practice*, **4**, pp. 615-623 (October 1996).
- [Sim02] D. Simon and T. Chia, "Kalman filtering with state equality constraints," *IEEE Transactions on Aerospace and Electronic Systems*, **38**(1), pp. 128-136 (January 2002).
- [Sim05a] D. Simon and D.L. Simon, "Aircraft turbofan engine health estimation using constrained Kalman filtering," *ASME Journal of Engineering for Gas Turbines and Power*, **127**(2), pp. 323-328 (April 2005).
- [Sim06a] D. Simon, "Reduced order Kalman filtering without model reduction," *Control and Intelligent Systems*, 2006.

- [Sim06b] D. Simon and D.L. Simon, "Constrained Kalman filtering via density function truncation for turbofan engine health estimation," submitted for publication.
- [Sim06c] D. Simon, "A game theory approach to constrained minimax state estimation," *IEEE Transactions on Signal Processing*, **54**(2), pp. 405-412 (February 2006).
- [Sim69] K. Simon and A. Stubberud, "Reduced order Kalman filter," *International Journal of Control*, **10**(5), pp. 501-509 (1969).
- [Sim98] J. Simonoff, *Smoothing Methods in Statistics*, Springer-Verlag, New York, 1998.
- [Sim82] C. Sims, "Reduced-order modelling and filtering," in C. Leondes (Ed.), *Control and Dynamic Systems*, Vol. 18, pp. 55-103, 1982.
- [Sin73] R. Singer and R. Sea, "New results in optimizing surveillance system tracking and data correlation performance in dense multitarget environment," *IEEE Transactions on Automatic Control*, pp. 571-582 (December 1973).
- [Sio96] G. Siouris, *An Engineering Approach to Optimal Control and Estimation Theory*, John Wiley & Sons, New York, 1996.
- [Ske98] R. Skelton, T. Iwasaki, and K. Grigoriadis, *A Unified Algebraic Approach to Linear Control Design*, Taylor & Francis, London, 1998.
- [Skl57] J. Sklansky, "Optimizing the dynamic parameters of a track-while-scan system," *RCA Review*, **18**, pp. 163-185 (June 1957).
- [Smi92] A. Smith and A. Gelfand, "Bayesian statistics without tears: A sampling-resampling perspective," *The American Statistician*, **46**(2), pp. 84-88, (May 1992).
- [Smi78] M. Smith and A. Roberts, "An exact equivalence between the discrete- and continuous-time formulations of the Kalman filter," *Mathematics and Computers in Simulation*, **20**(2), pp. 102-109 (June 1978).
- [Sne97] J. Snell, "A conversation with Joe Doob," *Statistical Science*, **12**(4), pp. 301-311 (November 1997).
- [Sob63] V. Sobolev, *A Treatise on Radiative Transfer*, Van Nostrand, Princeton, New Jersey, 1963.
- [Son95] Y. Song and J. Grizzle, "The extended Kalman filter as a local asymptotic observer for discrete-time nonlinear systems," *Journal of Mathematical Systems, Estimation, and Control*, **5**(1), pp. 59-78.
- [Soo65] T. Soong, "On *a priori* statistics in minimum-variance estimation problems," *ASME Journal of Basic Engineering*, **87D**, pp. 109-112 (March 1965).
- [Sor68] H. Sorenson and A. Stubberud, "Nonlinear filtering by approximation of the *a posteriori* density," *International Journal of Control*, **8**(1), pp. 33-51 (1968).
- [Sor70] H. Sorenson, "Least squares estimation: from Gauss to Kalman," *IEEE Spectrum*, **7**, pp. 63-68 (July 1970).
- [Sor71a] H. Sorenson and J. Sacks, "Recursive fading memory filtering," *Information Science*, **3**, pp. 101-119 (January 1971).
- [Sor71b] H. Sorenson and D. Alspach, "Recursive Bayesian estimation using Gaussian sums," *Automatica*, **7**(4), pp. 465-479 (July 1971).
- [Sor80] H. Sorenson, *Parameter Estimation*, Marcel Dekker, New York, 1980.
- [Sor85] H. Sorenson (ed.), *Kalman Filtering: Theory and Application*, IEEE Press, New York, 1985.
- [Sot99] G. Soto, E. Mendes, and A. Razek, "Reduced-order observers for rotor flux, rotor resistance and speed estimation for vector controlled induction motor

- drives using the extended Kalman filter technique," *IEE Proceedings - Electric Power Applications*, **146**(3), pp. 282-288 (May 1999).
- [Spa88] J. Spall (editor), *Bayesian Analysis of Time Series and Dynamic Models*, Marcel Dekker, New York, 1988.
- [Sri70] K. Srinivasan, "State estimation by orthogonal expansion of probability distributions," *IEEE Transactions on Automatic Control*, **AC-15**(1), pp. 3-10 (February 1970).
- [Ste68] E. Stear and A. Stubberud, "Optimal filtering for Gauss-Markov noise," *International Journal of Control*, **8**(2), pp. 123-130 (July 1968).
- [Ste52] P. Stein, "Some general theorems on iterants," *Journal of Research of the National Bureau of Standards*, **48**, pp. 82-82 (1952).
- [Ste94] R. Stengel, *Optimal Control and Estimation*, Dover, New York, 1994.
- [Sti86] S. Stigler, *The History of Statistics: The Measurement of Uncertainty Before 1900*, Cambridge, Massachusetts, Harvard University Press, 1986.
- [Sto94] A. Stoorvogel and J. Van Schuppen, "An H_∞ -parameter estimator and its interpretation," *IFAC Symposium on System Identification*, pp. 1223-1226 (July 1994).
- [Str90] P. Strobach, *Linear Prediction Theory*, Springer-Verlag, New York, 1990.
- [Swe59] P. Swerling, "First order error propagation in a stagewise smoothing procedure for satellite observations," *The Journal of the Astronautical Sciences*, **6**, pp. 46-52 (1959).
- [Swe63] P. Swerling, "Comment on 'A statistical optimizing navigation procedure for space flight'," *AIAA Journal*, **1**, p. 1968 (1963).
- [Swe71] P. Swerling, "Modern state estimation methods from the viewpoint of the method of least squares," *IEEE Transactions on Automatic Control*, **AC-16**, pp. 707-719 (1971).
- [Tap80] B. Tapley and J. Peters, "Sequential estimation algorithm using a continuous UDU^T covariance factorization," *Journal of Guidance and Control*, **3**(4), pp. 326-331 (July 1980).
- [The94a] Y. Theodor, U. Shaked, and C. de Souza, "A game theory approach to robust discrete-time H_∞ estimation," *IEEE Transactions on Signal Processing*, **42**(6), pp. 1486-1495 (June 1994).
- [The94b] Y. Theodor and U. Shaked, "Game theory approach to H_∞ -optimal discrete-time fixed-point and fixed-lag smoothing," *IEEE Transactions on Automatic Control*, **39**(9), pp. 1944-1948 (September 1994).
- [The96] Y. Theodor and U. Shaked, "Robust discrete-time minimum-variance filtering," *IEEE Transactions on Signal Processing*, **44**(2), pp. 181-189 (February 1996).
- [Thi61] H. Thiel and A. Goldberger, "On pure and mixed statistical estimation in economics," *International Economic Review*, **2**(1), pp. 65-78 (January 1961).
- [Tho94a] S. Thomopoulos and L. Zhang, "Decentralized filtering with random sampling and delay," *Information Sciences*, **81**, pp. 117-131 (1994).
- [Tho94b] S. Thomopoulos, "Decentralized filtering and control in the presence of delays: Discrete-time and continuous-time case," *Information Sciences*, **81**, pp. 133-153 (1994).
- [Tho77] C. Thornton and G. Bierman, "Gram-Schmidt algorithms for covariance propagation," *International Journal of Control*, **25**(2), pp. 243-260 (February 1977).

- [Tsa83] C. Tsai and L. Kurz, "An adaptive robustizing approach to Kalman filtering," *Automatica*, **19**(3), pp. 278-288 (May 1983).
- [Tse94] J. Tse, J. Bentsman, and N. Miller, "Minimax long range parameter estimation," *IEEE Conference on Decision and Control*, pp. 277-282, December 1994.
- [Urs80] B. Ursin, "Asymptotic convergence properties of the extended Kalman filter using filtered state estimates," *IEEE Transactions on Automatic Control*, **AC-25**(6), pp. 1207-1211 (December 1980).
- [Van01] R. Van der Merwe and E. Wan, "The square-root unscented Kalman filter for state and parameter estimation," *IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 3461-3464 (May 2001).
- [Vau70] D. Vaughan, "A nonrecursive algebraic solution for the discrete Riccati equation," *IEEE Transactions on Automatic Control*, **AC-15**(5), pp. 597-599 (October 1970).
- [Ver84] S. Verdu and H. Poor, "Minimax linear observers and regulators for stochastic systems with uncertain second-order statistics," *IEEE Transactions on Automatic Control*, **AC-29**(6), pp. 499-511 (June 1984).
- [Ver86] M. Verhaegen and P. Van Dooren, "Numerical aspects of different Kalman filter implementations," *IEEE Transactions on Automatic Control*, **AC-31**(10), pp. 907-917 (October 1986).
- [Ver95] M. Verlaan and A. Hemmink, "Reduced rank square root filters for large scale data assimilation problem," *Second International Symposium on Assimilation of Observations in Meteorology and Oceanography*, pp. 247-252 (1995).
- [Ver02] J. Vermaak, C. Andrieu, A. Doucet, and S. Godsill, "Particle methods for Bayesian modeling and enhancement of speech signals," *IEEE Transactions on Speech and Audio Processing*, **10**, pp. 173-185 (March 2002).
- [Wan01] E. Wan and R. van der Merwe, "The unscented Kalman filter," in *Kalman Filtering and Neural Networks*, S. Haykin (ed.), John Wiley & Sons, New York, 2001.
- [Wat22] G. Watson, *A Treatise on the Theory of Bessel Functions*, Cambridge University Press, Cambridge, United Kingdom, 1922.
- [Wei74] R. Weinstock, *Calculus of Variations*, Dover, New York, 1974.
- [Wel87] M. Welch, "A Kalman filter perspective," *The American Statistician*, **41**, pp. 90-91 (1987).
- [Wen92] W. Wen and H. Durrant-Whyte, "Model-based multi-sensor data fusion," *IEEE International Conference on Robotics and Automation*, pp. 1720-1726, May 1992.
- [Wes85] M. West, P. Harrison, and H. Mignon, "Dynamic generalized linear models and Bayesian forecasting," *Journal of the American Statistical Association*, **80**(389), pp. 73-97 (March 1985).
- [Wie56] N. Wiener, *I Am a Mathematician*, MIT Press, Cambridge, Massachusetts, 1956.
- [Wie64] N. Wiener, *Extrapolation, Interpolation, and Smoothing of Stationary Time Series*, MIT Press, Cambridge, Massachusetts, 1964.
- [Wil81] W. Willman, "Edgeworth expansions in state perturbation estimation," *IEEE Transactions on Automatic Control*, **AC-26**(2), pp. 493-498 (April 1981).
- [Wis69] R. Wishner, J. Tabaczynski, and M. Athans, "A comparison of three non-linear filters," *Automatica*, **5**(4), pp. 487-496 (July 1969).

- [Wis70] R. Wishner, R. Larson, and M. Athans, "Status of radar tracking algorithms," *Symposium on Nonlinear Estimation Theory and its Applications*, San Diego, pp. 32-54, 1970.
- [Wri97] S. Wright, *Primal-Dual Interior-Point Methods*, SIAM Press, Philadelphia, Pennsylvania, 1997.
- [Xie94] L. Xie, C. Soh, and C. de Souza, "Robust Kalman filtering for uncertain discrete-time systems," *IEEE Transactions on Automatic Control*, **39**(6), pp. 1310-1314 (June 1994).
- [Xie04] L. Xie, L. Lu, D. Zhang, and H. Zhang, "Improved robust H_2 and H_∞ filtering for uncertain discrete-time systems," *Automatica*, **40**(5), pp. 873-880 (May 2004).
- [Xu02] S. Xu and T. Chen, "Reduced-order H_∞ filtering for stochastic systems," *IEEE Transactions on Signal Processing*, **50**(12), pp. 2998-3007 (December 2002).
- [Yae91] I. Yaesh and U. Shaked, "A transfer function approach to the problems of discrete-time systems: H_∞ -optimal linear control and filtering," *IEEE Transactions on Automatic Control*, **36**(11), pp. 1264-1271 (November 1991).
- [Yae92] I. Yaesh and U. Shaked, "Game theory approach to state estimation of linear discrete-time processes and its relation to H_∞ -optimal estimation," *International Journal of Control*, **55**(6), pp. 1443-1452 (1992).
- [Yae04] I. Yaesh and U. Shaked, "Min-max Kalman filtering," *Systems and Control Letters*, **53**(3-4), pp. 217-228 (November 2004).
- [Yon80] K. Yonezawa, "Reduced-order Kalman filtering with incomplete observability," *Journal of Guidance and Control*, **3**(3), pp. 280-282 (June 1980).
- [Yoo04] M. Yoon, V. Ugrinovskii, and I. Petersen, "Robust finite horizon minimax filtering for discrete-time stochastic uncertain systems," *Systems and Control Letters*, **52**(2), pp. 99-112 (June 2004).
- [Zad50] L. Zadeh and J. Ragazzini, "An extension of Wiener's theory of prediction," *Journal of Applied Physics*, **21**, pp. 645-655 (July 1950).
- [Zar00] P. Zarchan and H. Musoff, *Fundamentals of Kalman Filtering: A Practical Approach*, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2000.
- [Zha05a] H. Zhang, L. Xie, Y. Chai Soh, and D. Zhang, " H_∞ fixed-lag smoothing for discrete linear time-varying systems," *Automatica*, **41**(5), pp. 839-846 (May 2005).
- [Zha95] X. Zhang, A. Heemink, and J. Van Eijkeren, "Performance robustness analysis of Kalman filter for linear discrete-time systems under plant and noise uncertainty," *International Journal of Systems Science*, **26**(2), pp. 257-275 (February 1995).
- [Zha05b] Y. Zhang, Y. Soh, and W. Chen, "Robust information filter for decentralized estimation," *Automatica*, **41**(12), pp. 2141-2146 (December 2005).

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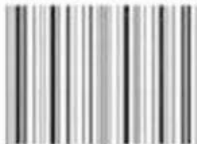
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